Unit: 6 Unit 5: Linear Equations
Topic: Scatter Plot and The Best Fit Line

Name: ____________
Date: ____________

Foundation (Jalapeno)

Q1. What is the purpose of a scatter plot?
(A) To show the relationship between two variables
(B) To display a single data point
(C) To create a line graph
(D) To make predictions
(E) To draw a circle

Q2. Which type of correlation is shown by the points $(1, 2)$, $(2, 4)$, $(3, 6)$?
(A) Negative
(B) None
(C) Positive
(D) Inverse
(E) Direct

Q3. What is the line of best fit used for?
(A) To find the median of the data
(B) To draw a vertical line
(C) To make predictions based on data trends
(D) To find the mode of the data
(E) To create a histogram

Q4. If the line of best fit has a positive slope, what can be said about the data?
(A) As x increases, y decreases
(B) As x increases, y increases
(C) The data has no correlation
(D) The data is constant
(E) The line is vertical

Q5. What does a strong positive correlation indicate?
(A) No relationship between the variables
(B) A weak relationship between the variables
(C) A moderate relationship between the variables
(D) A very strong relationship between the variables
(E) An inverse relationship between the variables

Q6. If a scatter plot shows a negative correlation, what happens to y as x increases?
(A) y increases
(B) y stays the same
(C) y decreases
(D) y becomes constant
(E) x and y switch places

Q7. What is the equation of the line of best fit if it passes through $(1, 2)$ and $(3, 4)$?
(A) $y = x + 1$
(B) $y = x - 1$
(C) $y = 2x - 1$
(D) $y = x + 2$
(E) $y = 2x + 1$

Q8. How can you determine if a line of best fit is a good fit for the data?
(A) By checking if the line is vertical
(B) By checking if the line is horizontal
(C) By checking how close the data points are to the line
(D) By counting the number of data points
(E) By finding the median of the data

Open Ended Questions (FRQ)

Q9. Create a scatter plot for the data points $(1, 3)$, $(2, 5)$, $(3, 7)$, and draw the line of best fit. Use the line to predict the value of y when $x = 4$.

Q10. The number of hours studied and the grade received on a test are related. If the data points are $(2, 80)$, $(4, 90)$, $(6, 95)$, find the equation of the line of best fit and use it to predict the grade if a student studies for 8 hours.

Chef's Tips

- Tip 1: Make sure students understand the concept of correlation and how it relates to the line of best fit.
- Tip 2: Encourage students to use the line of best fit to make predictions based on data trends.
- Tip 3: Remind students to check their work and ensure their calculations are accurate.